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Course : EDS

Practical 2

1)

Program :

```
l1 = []
```

```
while True:
```

```
    print("1. Add")
```

```
    print("2. Remove")
```

```
    print("3. Display")
```

```
    print("4. Quit")
```

```
    n = int(input("Enter choice: "))
```

```
    if n == 1:
```

```
        add = int(input("Integer: "))
```

```
        l1.append(add)
```

```
        print(f"List after adding: {l1}")
```

```
    elif n == 2:
```

```
        if len(l1) == 0:
```

```
            print("List is empty")
```

```
        elif len(l1) != 0:
```

```
            remove = int(input("Integer: "))
```

```
            if remove not in l1 :
```

```
                print("Element not found")
```

```
            else:
```

```
                l1.remove(remove)
```

```
                print(f"List after removing: {l1}")
```

```

elif n == 3:

    if len(l1) == 0:

        print("List is empty")

    else:

        print(l1)

elif n == 4:

    break

else:

    print("Invalid choice")

```

Output :

listOps.py

Average time: **0.408 s** (407.67 ms) | Maximum time: **0.582 s** (582.00 ms)

2 out of 2 shown test case(s) passed
1 out of 1 hidden test case(s) passed

Test case 1: 582 ms [Debug]

Expected output	Actual output
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 1	Enter choice: 1
Integer: 11	Integer: 11
List after adding: [11]	List after adding: [11]
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 1	Enter choice: 1
Integer: 25	Integer: 25
List after adding: [11, 25]	List after adding: [11, 25]
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 2	Enter choice: 2

listOps.py

Average time
0.408 s
407.67 ms

Maximum time
0.582 s
582.00 ms

2 out of 2 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Enter choice: 2
Integer: 26
Element not found
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 2
Integer: 11
List after removing: [25]
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 2
Integer: 25
List after removing: []
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 3

Enter choice: 2
Integer: 26
Element not found
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 2
Integer: 11
List after removing: [25]
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 2
Integer: 25
List after removing: []
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 3

listOps.py

Average time
0.408 s
407.67 ms

Maximum time
0.582 s
582.00 ms

2 out of 2 shown test case(s) passed

1 out of 1 hidden test case(s) passed

4. Quit
Enter choice: 3
List is empty
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 2
List is empty
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 8
Invalid choice
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 4

4. Quit
Enter choice: 3
List is empty
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 2
List is empty
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 8
Invalid choice
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 4

listOps.DV

Average time

0.408 s

407.67 ms

Maximum time

0.582 s

582.00 ms

2 out of 2 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 2 342 ms Debug

Expected output	Actual output
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 1	Enter choice: 1
Integer: 1	Integer: 1
List after adding: [1]	List after adding: [1]
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 1	Enter choice: 1
Integer: 2	Integer: 2
List after adding: [1, 2]	List after adding: [1, 2]
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 1	Enter choice: 1

listOps.DV

Average time

0.408 s

407.67 ms

Maximum time

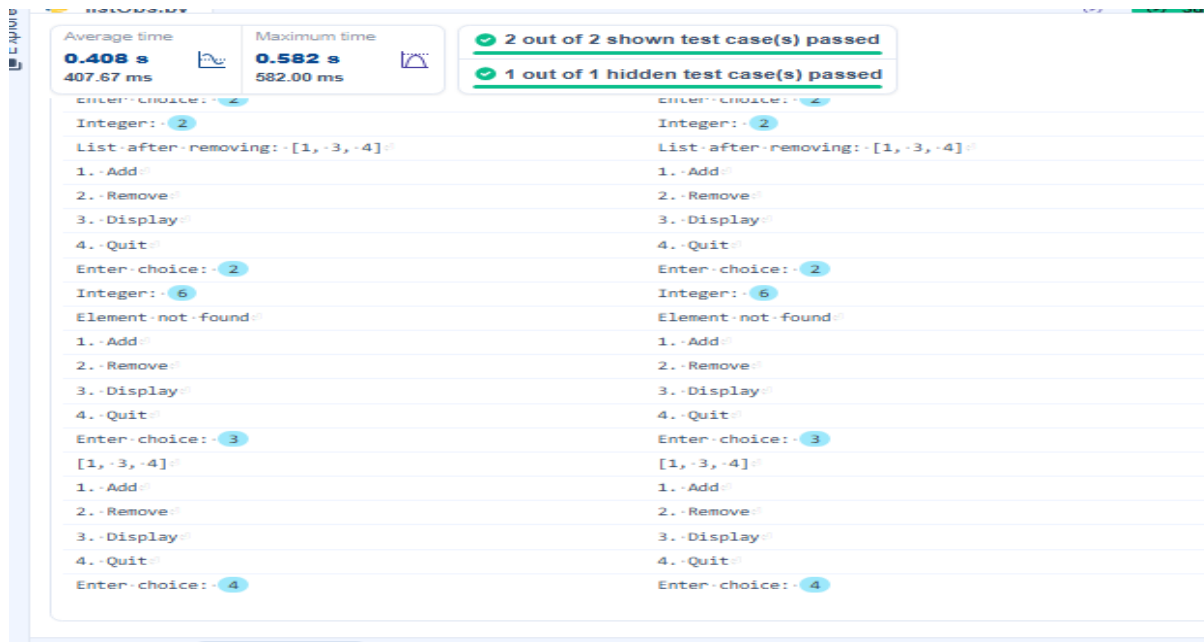
0.582 s

582.00 ms

2 out of 2 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Integer: 3	Integer: 3
List after adding: [1, 2, 3]	List after adding: [1, 2, 3]
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 1	Enter choice: 1
Integer: 4	Integer: 4
List after adding: [1, 2, 3, 4]	List after adding: [1, 2, 3, 4]
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 2	Enter choice: 2
Integer: 2	Integer: 2
List after removing: [1, 3, 4]	List after removing: [1, 3, 4]
1. Add	1. Add
2. Remove	2. Remove
3. Display	3. Display
4. Quit	4. Quit
Enter choice: 2	Enter choice: 2
Integer: 6	Integer: 6



2)

Program :

Initial dictionary with 10 predefined records

```
student = {
    1: "Amit",
    2: "Riya",
    3: "Kiran",
    4: "Neha",
    5: "Arjun",
    6: "Pooja",
    7: "Rahul",
    8: "Sneha",
    9: "Vikram",
    10: "Anjali"
}
```

```
# write your code here...
```

```
print("Original Dictionary:",student)
```

```
key1= int(input())
```

```
value1=input()
```

```
student.update({key1:value1})
```

```
print("After Insertion:",student)
```

```
key2=int(input())
```

```
value2=input()
```

```
if (key2 in student.keys()):
```

```
    student.update({key2:value2})
```

```
print("After Update:",student)
```

```
key3=int(input())
```

```
if (key3 in student.keys()):
```

```
    student.pop(key3)
```

```
print("After Deletion:",student)
```

```
print("Traversing Dictionary:")
```

```
for key,value in student.items():
```

```
    print(f"{key} : {value}")
```

Output :

Average time
0.070 s
70.00 ms

Maximum time
0.096 s
96.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 96 ms Debug

Expected output

Actual output

Original-Dictionary:--{1:-'Amit',-2:-'Riya',-3:-'Kiran',-4:-'Neha',-5:-'Arjun',-6:-'Pooja',-7:-'Rahul',-8:-'Sneha',-9:-'Vikram',-10:-'Anjali'}
11
Suresh
After-Insertion:--{1:-'Amit',-2:-'Riya',-3:-'Kiran',-4:-'Neha',-5:-'Arjun',-6:-'Pooja',-7:-'Rahul',-8:-'Sneha',-9:-'Vikram',-10:-'Anjali',-11:-'Suresh'}
3
Karthik
After-Update:--{1:-'Amit',-2:-'Riya',-3:-'Karthik',-4:-'Neha',-5:-'Arjun',-6:-'Pooja',-7:-'Rahul',-8:-'Sneha',-9:-'Vikram',-10:-'Anjali',-11:-'Suresh'}
5
After-Deletion:--{1:-'Amit',-2:-'Riya',-3:-'Karthik',-4:-'Neha',-6:-'Pooja',-7:-'Rahul',-8:-'Sneha',-9:-'Vikram',-10:-'Anjali',-11:-'Suresh'}
Traversing-Dictionary:
1:-Amit
2:-Riya
3:-Karthik
4:-Neha
6:-Pooja

Original-Dictionary:--{1:-'Amit',-2:-'Riya',-3:-'Kiran',-4:-'Neha',-5:-'Arjun',-6:-'Pooja',-7:-'Rahul',-8:-'Sneha',-9:-'Vikram',-10:-'Anjali'}
11
Suresh
After-Insertion:--{1:-'Amit',-2:-'Riya',-3:-'Kiran',-4:-'Neha',-5:-'Arjun',-6:-'Pooja',-7:-'Rahul',-8:-'Sneha',-9:-'Vikram',-10:-'Anjali',-11:-'Suresh'}
3
Karthik
After-Update:--{1:-'Amit',-2:-'Riya',-3:-'Karthik',-4:-'Neha',-5:-'Arjun',-6:-'Pooja',-7:-'Rahul',-8:-'Sneha',-9:-'Vikram',-10:-'Anjali',-11:-'Suresh'}
5
After-Deletion:--{1:-'Amit',-2:-'Riya',-3:-'Karthik',-4:-'Neha',-6:-'Pooja',-7:-'Rahul',-8:-'Sneha',-9:-'Vikram',-10:-'Anjali',-11:-'Suresh'}
Traversing-Dictionary:
1:-Amit
2:-Riya
3:-Karthik
4:-Neha
6:-Pooja

Average time
0.070 s
70.00 ms

Maximum time
0.096 s
96.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 2 74 ms Debug

Expected output

Actual output

Original-Dictionary:--{1:-'Amit',-2:-'Riya',-3:-'Kiran',-4:-'Neha',-5:-'Arjun',-6:-'Pooja',-7:-'Rahul',-8:-'Sneha',-9:-'Vikram',-10:-'Anjali'}
12
Meera
After-Insertion:--{1:-'Amit',-2:-'Riya',-3:-'Kiran',-4:-'Neha',-5:-'Arjun',-6:-'Pooja',-7:-'Rahul',-8:-'Sneha',-9:-'Vikram',-10:-'Anjali',-12:-'Meera'}
6
Divya
After-Update:--{1:-'Amit',-2:-'Riya',-3:-'Kiran',-4:-'Neha',-5:-'Arjun',-6:-'Divya',-7:-'Rahul',-8:-'Sneha',-9:-'Vikram',-10:-'Anjali',-12:-'Meera'}
1
After-Deletion:--{2:-'Riya',-3:-'Kiran',-4:-'Neha',-5:-'Arjun',-6:-'Divya',-7:-'Rahul',-8:-'Sneha',-9:-'Vikram',-10:-'Anjali',-12:-'Meera'}
Traversing-Dictionary:
2:-Riya
3:-Kiran
4:-Neha
5:-Arjun
6:-Divya

Original-Dictionary:--{1:-'Amit',-2:-'Riya',-3:-'Kiran',-4:-'Neha',-5:-'Arjun',-6:-'Pooja',-7:-'Rahul',-8:-'Sneha',-9:-'Vikram',-10:-'Anjali'}
12
Meera
After-Insertion:--{1:-'Amit',-2:-'Riya',-3:-'Kiran',-4:-'Neha',-5:-'Arjun',-6:-'Pooja',-7:-'Rahul',-8:-'Sneha',-9:-'Vikram',-10:-'Anjali',-12:-'Meera'}
6
Divya
After-Update:--{1:-'Amit',-2:-'Riya',-3:-'Kiran',-4:-'Neha',-5:-'Arjun',-6:-'Divya',-7:-'Rahul',-8:-'Sneha',-9:-'Vikram',-10:-'Anjali',-12:-'Meera'}
1
After-Deletion:--{2:-'Riya',-3:-'Kiran',-4:-'Neha',-5:-'Arjun',-6:-'Divya',-7:-'Rahul',-8:-'Sneha',-9:-'Vikram',-10:-'Anjali',-12:-'Meera'}
Traversing-Dictionary:
2:-Riya
3:-Kiran
4:-Neha
5:-Arjun
6:-Divya

3)

Program :

```
def linear_search(arr, key):  
    for i in range(len(arr)):  
        if arr[i] == key:  
            return i  
    return -1  
  
arr = list(map(int, input().split()))  
key = int(input())  
result = linear_search(arr, key)  
if result != -1:  
    print(result)  
else:  
    print("Not found")
```

Output :

The screenshot displays a code execution environment with the following details:

- Performance Metrics:**
 - Average time: 0.019 s (18.75 ms)
 - Maximum time: 0.025 s (25.00 ms)
- Test Results:**
 - 2 out of 2 shown test case(s) passed
 - 2 out of 2 hidden test case(s) passed
- Test Case 1 (25 ms):**
 - Expected output: 1 2 3 4 3 5 6
 - Actual output: 1 2 3 4 3 5 6
 - Input: 3
 - Output: 2
- Test Case 2 (19 ms):**
 - Expected output: 1 2 3 4 5 6 7 8 9 19
 - Actual output: 1 2 3 4 5 6 7 8 9 19
 - Input: 20
 - Output: Not - found

4)

Program :

```
heights = list(map(int, input().split()))
```

```
tallest_player = max(heights)
```

```
print(tallest_player)
```

Output :

The screenshot displays a code execution environment with the following components:

- Performance Metrics:**
 - Average time: 0.008 s (8.00 ms)
 - Maximum time: 0.011 s (11.00 ms)
- Test Results:**
 - 1 out of 1 shown test case(s) passed
 - 2 out of 2 hidden test case(s) passed
- Test Case Details:**
 - Test case 1: 11 ms, with a 'Debug' button and a menu icon.
- Output Comparison:**

Expected output	Actual output
171 169 185 156 174 191 186 190 187 172 160	171 169 185 156 174 191 186 190 187 172 160
191	191